

Unified AI-Native Product Specification

TATTVA COPILOT

Product Definition + Agent System + AI Evaluation + Operating Requirements

V1 — Story Development Copilot / Narrative Intelligence

September 2026

Document status	Value
Purpose	Unified implementation-ready specification for Product, Design, Engineering, AI/ML, Data, QA, Security and Leadership.
Primary source set	Business Requirements Document v1.0; Product Discovery Artifact v1.0; AI-Native Product Strategy v1.0; V1 Product Definition & Planning.
Source discipline	Source Requirement / Derived Requirement / Recommendation / Open Decision are explicitly distinguished.
Important source note	No standalone attachment titled “Agent Specifications” or “AI Evaluation Plan” was present in the uploaded set. Those sections are therefore synthesized from the source product, strategy, architecture and engineering material; they are not presented as independently approved source documents.
V1 boundary	Story-development intelligence and structured written Story Development Package. Visual generation, AI filmmaking, localization and full production operations remain future scope.

Status Legend

Label	Meaning
Source Requirement	Explicitly stated in the source artifacts.
Derived Requirement	Logically necessary to implement a source requirement or reconcile multiple source requirements.
Recommendation	Proposed design/operating choice added to close a gap; requires owner validation where material.
Open Decision	A decision that the source material explicitly leaves unresolved and that should not be silently decided by engineering.

1. Executive Summary

Tattava Copilot V1 is an AI-native Story Development and Project Intelligence system for entertainment teams. Its core value is not generic script generation. The product maintains a persistent representation of a creative project—story, characters, relationships, research, decisions, canon, versions and approvals—and uses that state to help people research, reason, develop, write, evaluate and maintain continuity.

[Source Requirement] The BRD defines V1 as the Story Development Copilot / Narrative Intelligence layer and establishes the Story Brain as the persistent project-intelligence core.

[Source Requirement] The strategy defines the AI-native distinction as project intelligence over isolated prompts, with AI generating, reasoning, validating and explaining while humans retain creative authority.

[Derived Requirement] Every AI capability must consume governed project context and must not maintain hidden project memory that can diverge from the Story Brain.

[Recommendation] Treat the product as one intelligent project-development system, not as a set of independent agents. Agent boundaries should be implementation contracts, not the primary UX.

The source artifacts converge on a Golden Workflow: Idea → Project/Intent → Story Brain → Research → Character Intelligence → Narrative Reasoning → Story Development → Format/Genre → Template → Writing → Screenplay → Dialogue → Canon/Continuity → Evaluation → Human Approval → Structured Story Development Package. Discovery, however, explicitly states that the complete capability chain is not yet validated as the smallest MVP. Therefore this specification distinguishes V1 capability architecture from MVP release scope.

The principal product hypothesis is that persistent, trustworthy, project-grounded intelligence can reduce context reconstruction, repetitive development work, research verification effort and continuity risk while improving the speed at which creative teams reach human-approved outputs.

2. Product Vision & Problem

2.1 Vision

[Source Requirement] Build the intelligence layer behind entertainment creation: every project remembers its story, research, characters, decisions, creative intent and dependencies, while AI continuously turns that intelligence into useful creative and production outcomes.

[Source Requirement] Long-term evolution: AI Assistant → AI Creative Copilot → AI Project Intelligence Layer → AI Production Operating System.

[Derived Requirement] V1 must create reusable structured project intelligence rather than disposable chat transcripts.

2.2 Problem Definition

Problem	Current consequence	Tattava response	Status
Fragmented project context	Information is spread across documents, people, notes, research and AI chats.	Persistent Project Intelligence / Story Brain.	Source Requirement
Repetitive development work	Research, synthesis, restructuring, rewriting and transformation are repeated.	Context-aware AI workflows and reusable project state.	Source Requirement
Research traceability	Creative decisions may be disconnected from evidence.	Source → evidence → finding → relevance → decision trace.	Source Requirement
Continuity risk	Characters, facts, timelines and logic drift across versions.	Canon + continuity detection + version-aware dependencies.	Source Requirement
Creative handoff friction	Research → story → writing → screenplay requires repeated translation.	Shared structured intelligence consumed by downstream capabilities.	Derived Requirement

The discovery artifact deliberately distinguishes known facts from hypotheses. It says the exact workflow, pain severity, largest time sink, current tools, AI usage, trust boundaries, switching triggers, ROI and exact P0 feature set remain unknown until user research and prototype validation. [filecite:turn0file2L790-L807](#)

The strategy therefore treats the persistent Project Intelligence Layer as a strategic hypothesis rather than a proven moat. [filecite:turn0file2L775-L807](#)

3. Users & Use Cases

Persona	Priority	Primary job	AI value	Authority
Story Developer / Story Editor	P0	Develop a coherent narrative.	Research, exploration, character intelligence, structure and continuity.	Human owns creative decisions.
Writer / Screenwriter	P0	Transform story concepts into written material.	Context-aware drafting, rewriting, expansion, refinement and screenplay transformation.	Writer remains author.
Creative Producer	P0/P1	Move projects toward approval.	Project state, evidence, readiness, options and structured outputs.	Approves according to organizational policy.
Director / Showrunner	P0/P1	Provide creative direction and high-level decisions.	Alternatives, consequences, consistency and decision support.	Creative authority.
Producer	P1	Track project progression and package readiness.	Structured project state and downstream readiness.	Policy-defined.
Production / Art / Specialist teams	P2/Future	Consume project intelligence downstream.	Future visual, production and operational intelligence.	Outside core V1.

4. AI-Native Product Principles

Principle	Operational interpretation
Project over prompt	The system understands the project before answering.
Context over generation	Optimize for correct, project-grounded output rather than maximum content volume.
Evidence before confidence	For factual research: Source → Evidence → Finding → Relevance.

Human authority	AI proposes; humans decide. Important state changes require explicit approval.
Traceability by default	Users can understand where information came from, why it changed, who approved it and what it affects.
One intelligence layer	All agents operate over governed Project Intelligence rather than isolated memory.
Validation is part of the product	Generation is followed by grounding, schema, policy, continuity or other relevant checks.
Progressive autonomy	Autonomy expands only where evidence and controls justify it.
Model agnostic	Durable value should reside above the underlying model/provider.
Workflow-first UX	AI appears inside creative workflows rather than as a generic chatbot destination.

The strategy explicitly defines the AI-native loop as understand → research/gather → generate → reason → check → explain → ask → revise → validate, with humans retaining creative authority. [filecite:turn0file2L94-L141](#)

5. Product Requirements

ID	Requirement	Class	Acceptance intent
PR-001	Create a project from idea, logline, concept, script/reference material, character ideas, themes or creative brief.	Source Requirement	Project and initial input persist; Story Brain initializes.
PR-002	Maintain persistent Story Brain containing project, story, characters, relationships, research, decisions, canon, versions, approvals and dependencies.	Source Requirement	Structured state is retrievable and versioned.
PR-003	Identify missing or ambiguous context before treating assumptions as project facts.	Source Requirement	Ambiguity is visible; user can answer or accept an explicit assumption.
PR-004	Conduct project-specific research with source/reference/finding/relevance provenance.	Source Requirement	No unsupported external claim is represented as verified evidence.
PR-005	Build character intelligence and relationship context.	Source Requirement	Downstream generation uses approved character state.
PR-006	Perform narrative reasoning over cause/effect, motivation, structure and logic.	Source Requirement	Issues are diagnostics/recommendations, not automatic creative decisions.
PR-007	Generate story alternatives and structured story artifacts.	Source Requirement	Candidate output is editable, rejectable, regenerable and approvable.
PR-008	Provide context-aware writing workflow.	Source Requirement	Output is grounded in current approved project state.
PR-009	Maintain canon and detect contradictions.	Source Requirement	Conflict evidence and affected entities are shown; human resolves.
PR-010	Evaluate project/output quality against explicit criteria.	Source Requirement	Evaluation produces strengths, gaps, risks and improvements; it does not approve.
PR-011	Preserve versions and approval history.	Source Requirement	Material changes create attributable versions.
PR-012	Assemble a structured Story Development Package.	Source Requirement	Package contains available approved artifacts and completeness diagnostics.
PR-013	Enforce project isolation, permissions and auditability.	Source Requirement	No unauthorized cross-project retrieval or mutation.
PR-014	Make AI runs observable and evaluable.	Derived Requirement	Run, model, context, tool, outcome, latency and cost are traceable.

6. Core User Journeys

Create Project

1. User submits idea/material
2. AI extracts intent/entities and detects ambiguity

3. System asks targeted questions where needed
4. Initial Story Brain is proposed
5. User edits/approves interpretation
6. Approved state becomes project intelligence

Research

1. User asks a project-specific question
2. System retrieves relevant project context
3. Research questions are generated
4. Sources are searched/evaluated
5. Findings are synthesized with provenance
6. User accepts/modifies/rejects
7. Approved finding becomes project intelligence

Story Development

1. Load Story Brain + research + character/canon state
2. User provides creative direction
3. Story Agent produces alternatives
4. AI explains rationale and constraints
5. User compares/edits
6. Candidate is approved or rejected
7. Approved artifact updates project state

Context-Aware Writing

1. User selects artifact/section
2. Context Resolver selects relevant project slice
3. Writing Agent drafts/rewrites
4. Context and risk signals are shown
5. User edits/regenerates/rejects/approves
6. New version is stored

Continuity

1. New content created
2. Relevant facts/entities are extracted
3. Canon is retrieved
4. System compares content against canon and dependencies
5. Conflict is shown with evidence and source/version
6. Human resolves/dismisses
7. Resolution becomes explicit project state

Final Package

1. Project reaches sufficient maturity
2. System collects approved artifacts

3. Completeness/quality checks run
4. Missing or weak areas are shown
5. Human approves package
6. Structured Story Development Package is generated

7. AI / Agent Architecture

The recommended architecture has five logical layers: Experience → Product Orchestration → Project Intelligence Layer → AI/Agent Layer → Model/Tool Layer. The Story Brain is the system of record; agent execution state and conversation context are temporary and non-authoritative.

Layer	Responsibilities
Experience	Project workspace, Story Brain, Research, Characters, Story, Writing, Screenplay, Canon, Evaluation, Versions and Final Package.
Orchestration / Policy	Intent routing, workflow state, permission enforcement, approval gates, version checks, retries, idempotency and job lifecycle.
Project Intelligence	Entities, relationships, evidence, decisions, canon, artifacts, versions, approvals and dependencies.
Agent Runtime	Agent contracts, context assembly, model calls, tool calls, structured output validation and telemetry.
Model / Tools	LLM providers, retrieval, web/search, parsers, embeddings and deterministic rules. Providers remain abstracted.

[Derived Requirement] Agents should never receive the entire Story Brain by default. A Context Resolver should assemble the minimum relevant, version-aware, project-scoped context package.

[Recommendation] Use a modular application plus asynchronous worker/job subsystem for V1 rather than prematurely splitting every agent into a microservice. The architecture should separate deployment boundaries from logical agent responsibilities.

[Derived Requirement] Every long-running AI operation needs a resumable job state, correlation ID, idempotency key, project/version context and explicit retry/failure state.

8. Agent Specifications

The following is the unified agent contract. It synthesizes the source product definition, architecture and engineering material; it should be treated as the implementation baseline, not as a separate previously approved Agent Specification.

Agent	Purpose	Context	Tools	Writes	Approval boundary
Intent	Interpret user goal and missing context.	User request + current project state.	Project retrieval.	Intent/unknowns candidate.	No creative approval; silently in canon.
Exploration	Generate meaningful directions/options.	Intent + Story Brain + relevant research.	Retrieval; research if authorized.	Candidate exploration.	Human ch direction.
Research	Find reliable evidence for project questions.	Project + research question.	Search/retrieval/parsing.	Research plan/evidence records.	Insufficient evidence explicit; no fabricated evidence.
Character Intelligence	Maintain structured character model.	Approved character-related state.	Project retrieval.	Character/relationship candidates.	Human re ambiguity.

Narrative Reasoning	Assess causal and structural logic.	Story, characters, canon, research.	Structured project query + rules.	Diagnostics.	Findings and recommendations; critical block policy is a Decision.
Story	Develop premise, synopsis, structure, themes, conflict and beats.	Story Brain + research + characters + canon.	Project retrieval.	Candidate story artifacts/versions.	Hard human approval to downstream gate.
Format/Genre	Advise format/genre choices.	Project/story context.	Template catalog.	Recommendation/configuration.	Human selection for final form.
Template	Apply a deterministic development structure.	Selected format/genre.	Template registry.	Template configuration.	Deterministic; can change selection.
Writing	Draft, rewrite, expand, shorten and refine.	Relevant approved story, characters, canon, research.	Project retrieval; writing tools.	Candidate artifact/version.	Never silently overwrite content.
Screenplay	Transform approved story into screenplay structure.	Approved story + format/template.	Project retrieval.	Candidate screenplay.	Requires human approval; review before further progression.
Dialogue	Generate context-aware dialogue.	Character + scene + tone + story state.	Project retrieval.	Candidate dialogue.	Human review; voice is not automatic authority.
Story Doctor	Diagnose and propose targeted repairs.	Story/screenplay/dialogue + project context.	Retrieval + rules.	Diagnostic/fix suggestions.	Does not autonomously rewrite approved state.
Canon/Continuity	Detect contradictions against established facts.	Canon + current artifact + versions + dependencies.	Graph/rules/query.	Findings and explicit resolutions.	Never automatically resolves conflicts.
Evaluation	Assess readiness and quality.	Approved artifacts + rubric.	Deterministic checks + evaluator model.	Evaluation report.	Evaluation approval.
Impact / Dependency	Explain downstream effects of upstream changes.	Dependency graph + changed entity/version.	Graph + rules + LLM explanation.	Impact report/staleness markers.	P1/V1-late explicitly promoted.

9. Agent Roles, Responsibilities & Boundaries

[Derived Requirement] Every agent must declare: ID/version, purpose, trigger, inputs, context requirements, tools, model policy, output schema, validation rules, allowed writes, emitted events, approval requirement, failure modes, retry policy, evaluation criteria and telemetry.

[Source Requirement] No agent may silently mutate approved canon or project state.

[Derived Requirement] All agent outputs should first be treated as candidate results. Only validated and approved mutations can become authoritative project intelligence.

The most important boundary is between cognitive work and state mutation. LLMs can reason, propose, summarize and generate. Deterministic services should enforce identity, authorization, project isolation, approval state, version creation, audit logging, package assembly and state-transition rules.

10. Tools, Data & Integrations

Data / tool	Role	Trust / governance
Project Intelligence / Story Brain	Authoritative project state.	Project-scoped ACL; versioned; attributable; canonical state separated from candidates.
Research sources	External evidence.	Source provenance, source quality, citation validation, conflict handling.
Search / web	Research retrieval.	Tool permission, source policy, untrusted-content handling.
Document parser	Extract uploaded project material.	Treat uploaded instructions as untrusted content; quarantine malicious files.
Vector / semantic retrieval	Find relevant project context.	Never substitute vector similarity for authorization or canonical state.
Model gateway	Provider abstraction and model routing.	No hard-coded provider dependency; approved model list and data policy are Open Decisions.
Evaluation datasets	Benchmark and regression testing.	Versioned datasets and judgments; restricted access.
Observability	Logs, traces, metrics, costs and evaluation telemetry.	Minimize sensitive creative content in raw telemetry.

[Open Decision] Which model providers are approved, what data may be sent to them, retention/deletion rules, and the maximum acceptable project/month and per-run AI cost.

[Open Decision] Which external research sources are permitted for production use and what source hierarchy governs conflicts.

11. Human-in-the-Loop Requirements

The operating loop is: AI proposes → human reviews → human edits/accepts/rejects → system records decision → approved state becomes project intelligence → future AI uses the approved state.

State	Meaning	Can become canonical?
AI Generated	Raw model/tool output.	No
Candidate	Validated proposal awaiting human decision.	No
Human Edited	User-modified candidate.	No, until approved.
Human Approved	Explicitly accepted project artifact/state.	Yes
Canonical Project State	Authoritative state used by downstream agents.	Yes
Superseded	Older approved version retained for provenance.	No for new generation unless selected by explicit version action.

[Derived Requirement] Story approval is a hard workflow gate before downstream screenplay generation where the source blueprint requires it.

[Source Requirement] Canon conflicts are always routed to a human decision; the system must never auto-resolve them.

[Open Decision] Exact approval hierarchy by organization, project stage and artifact type is not specified in the source material.

12. AI UX & Interaction Model

[Source Requirement] The product should not primarily resemble ChatGPT + sidebar + Generate. AI should appear inside workflows.

Recommended interaction pattern:

- The user starts from a project object or artifact, not a blank chat.
- The UI shows the current approved state and relevant context before proposing changes.
- AI suggestions are visually distinct from canonical content.
- Every meaningful suggestion can expose source/context, rationale, uncertainty and affected entities.
- Actions are outcome-oriented: Research this, Explore three directions, Rewrite this scene using approved canon, Check continuity, Explain this conflict.
- Approval is explicit and local to the artifact/state being changed.
- Material changes create versions and preserve the prior state.
- When context is insufficient, the AI asks targeted questions instead of filling gaps invisibly.

Example: on a Character page, the user sees current motivation, relationships and canon; Tattava can surface a possible motivation conflict, relevant research and several development directions with Accept / Edit / Reject controls.

13. System Behavior & Decision Flows

13.1 Context resolution

1. Receive user intent and current project/version.
2. Authorize project access.
3. Identify required entities/artifacts for the selected capability.
4. Retrieve canonical facts, relevant approved artifacts, recent changes and dependencies.
5. Exclude unrelated or unauthorized project data.
6. Build a minimal context package.
7. Execute the agent.
8. Validate output.
9. Return candidate result plus provenance and uncertainty.
10. Commit only after required approval/state transition.

13.2 Version and dependency behavior

Every generated artifact should record the Story Brain entities and versions it consumed. If an upstream entity changes, dependent artifacts become stale rather than silently remaining apparently valid. The user can accept, edit or regenerate the dependent artifact, after which the relevant quality checks rerun.

[Recommendation] Implement explicit dependency edges for at least Character → Story Beat → Scene → Dialogue and Canon Fact → dependent artifacts in the first architecture increment, even if advanced impact intelligence remains P1.

13.3 Workflow state

Recommended job state: CREATED → READY → QUEUED → RUNNING → WAITING_FOR_APPROVAL / COMPLETED / FAILED_RETRYABLE / FAILED_FINAL / CANCELLED / STALE.

[Derived Requirement] Long-running work must be asynchronous, resumable and idempotent. An old job must not mutate a newer project version without explicit reconciliation.

14. Safety, Security & Guardrails

Control	Requirement	Status
Project isolation	No cross-project retrieval or mutation.	Source Requirement
RBAC / authorization	View/edit/generate/approve permissions are enforced server-side.	Source + Derived
IP protection	Creative assets and unreleased scripts are treated as sensitive IP.	Source Requirement
Approval gates	Critical creative state transitions require human approval.	Source Requirement
Prompt/document injection	Uploaded/retrieved content is untrusted and must not execute instructions.	Derived Requirement

Research integrity	No credible evidence → explicitly state insufficient evidence.	Source/Derived
Conflicting sources	Surface conflict rather than silently choosing.	Derived
Auditability	Important mutations are attributable to actor, agent/run and version.	Source Requirement
Secrets	No credentials or provider secrets in prompts, outputs or user-visible traces.	Recommendation
Data retention	Retention/deletion and model-provider data handling are policy-controlled.	Open Decision

15. AI Evaluation Framework

Evaluation must be designed before launch, not after implementation. The source product definition calls for benchmark datasets covering research source/citation correctness and unsupported claims; story project grounding and user acceptance; character consistency; continuity contradiction precision/recall/false positives; and overall time-to-usable-output/time-to-approved-package.

[Source Requirement] Separate the internal AI evaluation harness from the user-facing Evaluation Agent. The harness evaluates system/agent quality; the Evaluation Agent evaluates a particular creative project against a rubric.

[Derived Requirement] Every agent must have at least one offline benchmark, one online/product metric and one human-quality signal before it is considered release-ready.

16. Evaluation Metrics & Success Criteria

Dimension	Metric	Definition	Target status
Research	Source correctness	% of cited sources that actually support the claim.	Recommendation: threshold set from pilot benchmark.
Research	Citation completeness	% of externally verifiable claims with appropriate evidence/citation.	Recommendation.
Research	Unsupported claim rate	% of factual claims without adequate evidence.	Recommendation: near-zero for verified research mode.
Research	Source relevance	Human/benchmark judgment that sources answer the project question.	Recommendation.
Grounding	Project grounding	% of generated content consistent with required approved context.	Recommendation.
Grounding	Context omission rate	% of required context elements omitted or contradicted.	Recommendation.
Story	Human acceptance rate	Accepted candidates / reviewed candidates.	Baseline first; target after pilot.
Story	Edit distance	Magnitude of user edits required before acceptance.	Baseline first; lower is better.
Writing	Regeneration rate	Regenerations per accepted output.	Baseline first.
Character	Consistency	Correctness of character attributes, motivations, relationships and voice constraints.	Benchmark threshold required.
Continuity	Precision	True contradictions / all flagged contradictions.	Benchmark threshold required.
Continuity	Recall	True contradictions detected / all benchmark contradictions.	Benchmark threshold required.
Continuity	False-positive rate	Non-contradictions incorrectly flagged.	Keep low enough to preserve trust.
Evaluation	Rubric agreement	Agreement between evaluator output and expert judgment.	Benchmark required.
Product	Time to usable output	Elapsed time from task start to artifact usable by the professional.	Baseline and improvement target.
Product	Time to approved package	Elapsed time from accepted project input to human-approved package.	North-star V1 proxy.
Trust	Verification rate	Share of research outputs users verify or inspect.	Track, do not optimize blindly.
Adoption	Approved milestone rate	% projects reaching a human-approved milestone.	Core product signal.

The strategy explicitly warns against optimizing for number of generations, prompts or words; it recommends measuring projects moving meaningfully forward, including time saved, decision quality, continuity, trust and approved outputs.

filecite\turnOfile2\L759-L774

17. Test Sets, Scenarios & Edge Cases

Test family	Representative cases	Pass condition
Research grounding	Correct source, irrelevant source, conflicting sources, no credible source, broken URL.	Evidence is correctly attributed; uncertainty is explicit; no fabricated source.
Project grounding	New scene conflicts with approved motivation; writing request omits a known canon fact.	Output uses relevant approved context or asks for clarification.
Continuity	Character age 32 → 35; timeline inconsistency; relationship conflict; world-rule violation.	Conflict includes both evidence sides, affected entity and version.
Versioning	Two edits to same artifact; stale job; restore old version.	No silent overwrite; conflict is explicit.
Approval	Attempt screenplay generation before story approval.	Server-side gate prevents unauthorized progression.
Security	Cross-project retrieval; role escalation; unauthorized export.	Request denied and audited.
Prompt injection	Uploaded document contains instructions to reveal secrets or alter canon.	Content treated as data, not executable instruction.
Reliability	Model timeout, worker termination, provider outage.	Retryable failure preserves state; no duplicate mutation.
Malformed output	Agent returns invalid schema.	Output rejected; no Story Brain mutation.
Context overload	Very large project with many versions.	Scoped retrieval selects relevant context rather than full dump.
User behavior	Repeated regenerate clicks.	Idempotency/deduplication prevents duplicate committed changes.

18. Monitoring & Continuous Improvement

Continuous evaluation is part of the product operating model. Every model/prompt/agent version change should be regression-tested against the benchmark suite. During pilot, human sampling should inspect real outputs. Product signals such as rejection, regeneration and override rates should be interpreted as quality signals, not standalone measures of quality.

Monitoring layer	Signals
Agent	Latency, token usage, model, tool calls, validation failures, retry count, failure type.
Workflow	Stage completion, queue time, approval waits, stale jobs, retries, end-to-end time.
Quality	Grounding, citation, contradiction, acceptance, edit and evaluator agreement metrics.
Product	Projects created, returning projects, Story Brain activation, approved milestones, package completion.
Cost	Cost per run, agent, workflow and project; regeneration cost; research/search cost.
Security	Denied requests, cross-project attempts, privilege escalation attempts, suspicious tool usage.
User feedback	Feedback tied to artifact version and AI run wherever possible.

[Recommendation] Maintain a model/prompt release gate: offline benchmark passes → targeted regression → shadow/pilot comparison → controlled rollout → production monitoring → rollback if critical quality/safety thresholds regress.

19. Failure Handling & Escalation

Failure	Required behavior	Escalation
Insufficient research	State insufficient evidence; preserve project state; suggest further research.	User decides whether to continue.
Conflicting sources	Show conflict and provenance; do not silently select.	Human research/creative decision.
Invalid AI output	Reject mutation; retry/repair if safe.	Engineering telemetry if repeated.

Provider timeout	Retry according to policy; preserve job state.	Escalate after retry budget exhausted.
Stale context	Mark job/artifact stale; require reconciliation.	User or workflow manager.
Canon conflict	Present evidence and affected artifacts.	Human resolution required.
Permission revoked mid-run	Prevent result exposure/commit.	Security/audit event.
Storage failure	Preserve previous canonical state; make job recoverable.	Engineering incident if persistent.
Evaluation failure	Project remains usable but cannot pass final quality gate.	Human review / product owner.

20. Technical Requirements

Area	Requirement
Architecture	Modular application + asynchronous job/worker model is recommended for V1; physical service decomposition remains an engineering decision.
State	Canonical project state, artifacts, versions, approvals, events and transient agent runs are distinct.
AI runtime	Standard agent contract; structured output; model gateway/provider abstraction.
Context	Version-aware, project-scoped, minimal retrieval.
Workflow	State machine, idempotency, retries, cancellation, resumability and stale-job protection.
Concurrency	Explicit conflict policy; no silent loss of edits.
API	Project, Story Brain, Artifact, Workflow, Agent and Evaluation responsibilities; protocol is an Open Decision.
Storage	Persistent structured state plus object/file storage as required; concrete database/provider remains an Engineering Decision.
Observability	Logs, metrics, traces, evaluation records and cost telemetry.
Testing	Unit, integration, API, end-to-end, security, performance and failure-injection testing.

21. Non-Functional Requirements

NFR	Requirement	Target
Security	Project isolation, RBAC, secure storage, controlled AI context, auditability.	Must.
Reliability	No loss of canonical state; recoverable async jobs.	Must.
Correctness	Invalid/unsupported AI output cannot mutate canonical state.	Must.
Traceability	Human/agent/source/version provenance for important changes.	Must.
Performance	Interactive local actions should be responsive; long AI work should be asynchronous.	Exact numeric thresholds TBD.
Scalability	Support pilot concurrency and independent worker scaling where required.	Load profile TBD.
Cost	Measure and control model/tool cost at run, agent, workflow and project levels.	Budget TBD.
Privacy/IP	Sensitive creative material must not cross project or provider boundaries without policy.	Must.
Usability	Creative professionals should not need technical expertise to operate core workflows.	Must.
Maintainability	Agent contracts, schemas, prompts/policies and benchmark versions must be versioned.	Derived requirement.

22. Analytics & Product Metrics

North-star V1 proxy: Time from accepted project input to Human-Approved Story Development Package.

Metric group	Metrics
Efficiency	Research turnaround; time to first useful artifact; time per iteration; context reconstruction time; repetitive work saved.
Quality	Human acceptance; correction rate; canon consistency; contradiction detection; missed contradictions; research verification.
Trust	AI rejection; verification; approval; source interaction.

Adoption	Projects created; returning projects; active projects; Story Brain activation; approved output; workflow completion.
Readiness	Story package completeness; character completeness; research completeness; canon completeness; structured metadata completeness; downstream reuse.

[Open Decision] The source artifacts do not specify numerical baselines or release thresholds. Do not invent them in engineering tickets. Establish them through benchmark construction, pilot measurement and management approval.

23. MVP vs Future Roadmap

Capability	V1 architecture	MVP recommendation	Future
Project creation	Core	Must	—
Story Brain	Core	Must	Cross-stage intelligence
Research	Core	Must	Advanced source intelligence
Character intelligence	Core	Must	Deep character/world intelligence
Narrative reasoning	Core	Lightweight / validated depth	Advanced reasoning
Story development	Core	Must	—
Writing interface	Core	Must	Collaboration/branching
Canon/continuity	Core	Basic	Advanced dependency intelligence
Evaluation	Core	Basic + benchmark	Advanced quality/diagnostics
Format/genre	V1 capability	Lightweight	Deeper advisory intelligence
Template	V1 capability	Lightweight	Expanded registry
Screenplay	V1 capability	Depth is Open Decision	Deep screenplay intelligence
Dialogue	V1 capability	Depth is Open Decision	Advanced dialogue/voice intelligence
Story Doctor	V1 capability	P1 / V1-later unless promoted	Full diagnostic/repair
Dependency/Impact	V1-later	P1	Production consequence intelligence
Visual development	Future	No	V2
Image/video generation	Future	No	V2/V3
Production planning	Future	No	Layer 2
Localization	Future	No	V4
Business/management OS	Future	No	MVV OS

The most important scope conflict is explicit: the BRD defines a broad V1 capability pipeline, while Discovery says the smallest useful MVP is not yet validated. The product definition resolves this by separating V1 product architecture from MVP release scope. [filecite]turn0file3[L203-L221]

24. Risks, Dependencies & Open Questions

Item	Type	Why it matters	Required resolution
Persistent Project Intelligence creates real user value	Hypothesis	Central strategic differentiator.	User interviews + prototype + task measurement.
Users will change workflow	Hypothesis	Adoption depends on behavior change.	Workflow observation + switching interviews.
Research traceability increases trust	Hypothesis	Trust is central to evidence-backed AI.	Controlled research experiment.
Continuity is worth solving	Hypothesis	May be less painful than assumed.	Task-based benchmark + user study.
Exact MVP boundary	Open Decision	Controls engineering scope.	Prototype validation + management decision.
Screenplay P0 depth	Open Decision	Can materially expand V1 complexity.	Pilot objective + management decision.
Dialogue P0 depth	Open Decision	Voice/quality evaluation is difficult.	Pilot evidence.

Story Doctor P0/P1	Open Decision	Overlap with evaluation and narrative reasoning.	Define differentiated job and metric.
Approval hierarchy	Open Decision	Required for authoritative state.	Management/product policy.
Approved canon definition	Open Decision	Continuity depends on authority semantics.	Data/state policy.
Model providers	Open Decision	Security, quality and cost implications.	Architecture/management decision.
External research source policy	Open Decision	Trust and licensing implications.	Research policy.
Data retention/provider data policy	Open Decision	Sensitive IP.	Security/legal decision.
Project/version schema	Open Decision	Foundation for Story Brain.	Architecture decision.
Evaluation benchmark	Open Decision	Release quality cannot be controlled without it.	AI evaluation owner + domain experts.

25. Implementation Recommendations

1. Freeze the Story Brain entity/event/artifact/dependency model before implementing a large agent surface.
2. Define canonical-state semantics: AI proposal → candidate → human edited → human approved → canonical.
3. Build one vertical slice end-to-end before implementing every capability deeply.
4. Implement the Context Resolver early; context quality is a core product capability, not an optimization.
5. Implement Research as structured evidence, not as a text-generation endpoint.
6. Build Canon/Continuity into the workflow from the start rather than as a final QA button.
7. Create the evaluation harness and benchmark datasets before launch.
8. Keep the user experience outcome-oriented; users should not need to know which agent runs.
9. Enforce approval gates on the server, not only in UI.
10. Instrument every AI run for latency, model, cost, tool usage, context identifiers, validation and outcome.
11. Treat uploaded documents and retrieved web content as untrusted data for prompt-injection purposes.
12. Protect against stale writes and duplicate mutations through version checks and idempotency.
13. Establish pilot baselines before setting aggressive numerical quality thresholds.
14. Use human expert review to calibrate creative-quality evaluations rather than pretending one scalar score captures story quality.
15. Keep future extensibility in the data contracts, not by implementing future production features prematurely.

26. Appendix — Source-to-Requirement Traceability

Requirement area	BRD	Discovery	Strategy	Product Definition	Unified interpretation
V1 purpose	Story Development Copilot	Needs validation	Golden Workflow	Story Development + Project Intelligence	V1 capability architecture; MVP remains validated subset.
Story Brain	Persistent project core	Core hypothesis	Strategic differentiator hypothesis	System-of-record	Central product object.
Research	Credible + traceable	Trust hypothesis	Evidence-first	Structured provenance	First-class intelligence system.
Human control	Human-in-the-loop	Human-controlled AI hypothesis	Human authority	Approval state machine	Authoritative design principle.
Continuity	Canon + contradiction checks	High-value hypothesis	Workflow capability	Continuous checking	Core V1 capability; advanced impact P1.

Writing	Collaborative writer + AI	Context-aware generation hypothesis	Context over generation	Grounded writer interface	Core V1 workflow.
Evaluation	Structured quality pass	Benchmark needs validation	Quality + approved output	Evaluation agent + harness	Two distinct evaluation layers.
Screenplay	V1 capability	MVP uncertain	Golden workflow	Should/P0 decision	Scope depth remains Open Decision.
Dialogue	V1 capability	MVP uncertain	Golden workflow	Should	Scope depth remains Open Decision.
Story Doctor	V1 capability	MVP uncertain	Phase 1 focus excludes it	P1 / V1-later	Do not pull into MVP without decision.
Security	Secure IP / access	Trust concerns	Project isolation	RBAC + controlled retrieval	Foundational V1 requirement.
Metrics	Time / development / adoption	Efficiency / quality / trust / adoption	North-star hierarchy	Time to approved package	Outcome-based KPI system.

27. Final Operating Contract

Tattava V1 should be considered ready to pilot only when the following statement is true:

A creative professional can create a project from a real idea or source material; Tattava establishes structured project intelligence; research is traceable and does not fabricate evidence; AI outputs are grounded in approved project context; material creative changes are reviewable and versioned; continuity conflicts are surfaced with evidence rather than silently resolved; critical workflow transitions are human-approved; the complete journey is observable and recoverable; and the system can demonstrate measurable quality on representative benchmark projects.

[Source Requirement] The source artifacts explicitly position the immediate success condition as moving from accepted project input to an approved, researched, structured Story Development Package, while preserving human creative authority.

[Recommendation] Treat this as the V1 release contract and make every engineering epic, agent, evaluation suite and UX workflow traceable to it.

Source Notes

Primary source artifacts consulted: Business Requirements Document (BRD) v1.0; Product Discovery Artifact v1.0; Tattava Copilot — AI-Native Product Strategy v1.0; Tattava Copilot — V1 Product Definition & Planning.

Supporting implementation consistency was checked against project-library architecture/engineering artifacts where they clarify agent contracts, asynchronous workflow, versioning, evaluation and security. These supporting artifacts are not silently treated as higher authority than the primary product/strategy/discovery sources.

Key source excerpts: the strategy describes the persistent intelligence layer and human-controlled AI model.

“The product definition distinguishes the broad BRD V1 capability set from the not-yet-validated MVP boundary. The discovery artifact explicitly separates known facts, working hypotheses and unknowns.”